Week 85: Drawing machines

Adult guide built from the problem files · Unpiloted; no pantograph has been built or rehearsed.

This week's adult guide is week-85-facilitator.pdf, whose source is editable/guide/facilitator.tex. These problem files hold each problem's one-sentence idea, not its answers.

Problems and routes

Problem	2–5	4–5	route
Copy three paths	P1		first
Steer P through dots and round an outline	P2		first
A rule for the distances	P3		first
Keep the copy inside a box	P4		first
Why P is twice as far as T		P1	return
Area of a copy		P2	return
Swap the tips: half-size copies		P3	return
Move the anchor		P4	return

First route. Grades 2–5 P1–P4; Grades 4–5 none. **Return visits.** Grades 2–5 none; Grades 4–5 P1–P4.

Key, one entry per problem

Copy three paths

Grades 2–5 P1 · first route

Idea. The pencil P draws a copy of T's path twice as large, with the same shape and direction.

Steer P through dots and round an outline

Grades 2–5 P2 · first route

Idea. To make P reach a point, T goes halfway there from O; T's paths come out half size.

A rule for the distances

Grades 2–5 P3 · first route

Idea. In every open pose O, T and P lie on one line with P twice as far from O as T.

Keep the copy inside a box

Grades 2–5 P4 · first route

Idea. The copy stays inside a box exactly when T stays inside the half-size box, half as far from O.

Why P is twice as far as T

Grades 4–5 P1 · return route

Idea. XT is parallel to QP and half as long, so O, T, P are collinear and OP = 2 OT in every open pose.

Area of a copy

Grades 4–5 P2 · return route

Idea. Doubling every length multiplies area by four.

Swap the tips: half-size copies

Grades 4–5 P3 · return route

Idea. Swapping the tips gives the inverse dilation by 1/2 about the same O, so the target's position matters as well as its size.

Move the anchor

Grades 4–5 P4 · return route

Idea. Moving the anchor by d moves the copy by -d; moving the input by d/2 puts the copy back.